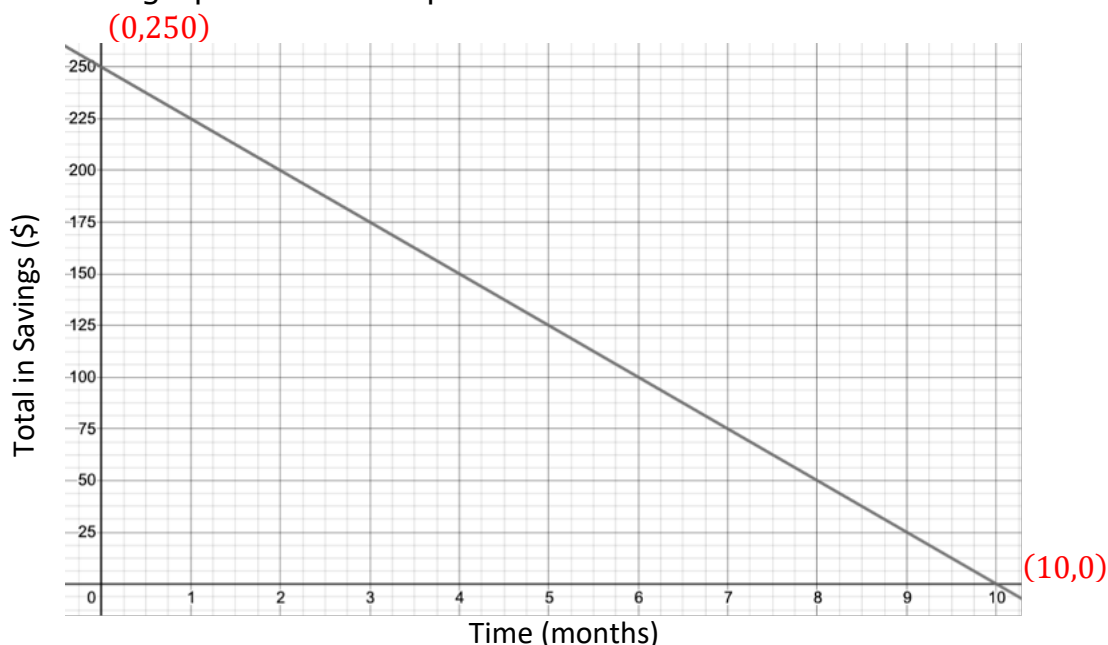


Grade 8
Unit 7
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

The graph represents the total in Kayla's savings account, y , over the amount of time, x , in months. Use the graph to answer questions 1–7.



1. What is the slope of the line?

$$\frac{250-0}{0-10} = \frac{250}{-10} = -25$$

2. What is the y -intercept of the line?

250

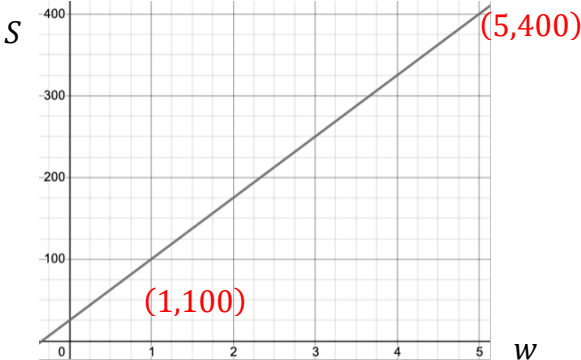
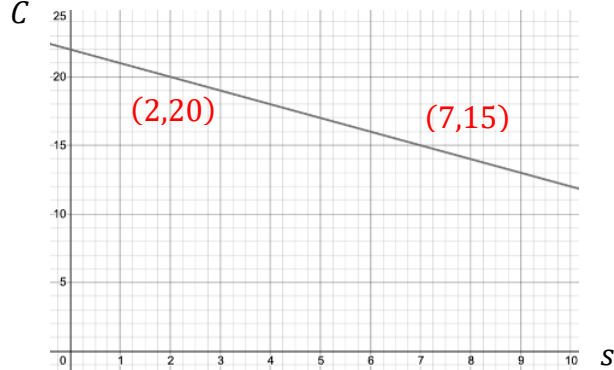
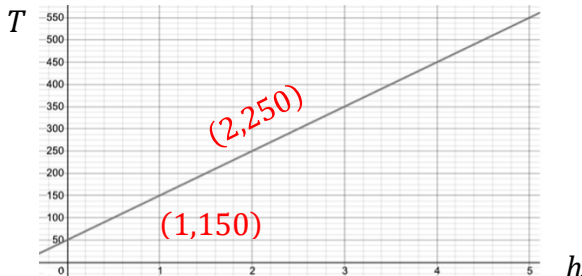
Use the graph to match each key feature with its interpretation.

Key Feature	Interpretation
3. x -intercept C	A. Kayla withdraws \$25 per month from her account.
4. y -intercept B	B. Kayla started with \$250 in her account.
5. Slope A	C. After 10 months Kayla's account is empty.
6. sign of the slope D	D. Kayla's savings account is decreasing.

7. Explain why the x -intercept does or does not make sense in this context.

It does because it represents how many months of making withdrawals before she ran out of money.

Determine the slope and y-intercept of each graph and interpret their meaning:

Graph	Slope and Interpretation	y-intercept and Interpretation
8. Sam is trying to save money to buy a new phone. The graph represents the relationship between the amount saved, S, and the number of weeks, w.  The graph shows a line on a coordinate plane where the horizontal axis is labeled w (weeks) and the vertical axis is labeled S (amount saved). The line starts at the y-intercept (0, 25) and passes through points (1, 100) and (5, 400), which are labeled in red.	$\frac{400-100}{5-1} = \frac{300}{4} =$ $\\$75/1$ week Sam saves \$75 per week.	(0,25) Sam started w/ \$25.
9. Trae and his friends are sharing a family size bag of chips. The amount left in the chip bag, C, in ounces is represented in the graph, with number of servings eaten, s.  The graph shows a line on a coordinate plane where the horizontal axis is labeled s (servings eaten) and the vertical axis is labeled C (amount left in ounces). The line starts at the y-intercept (0, 22) and passes through points (2, 20) and (7, 15), which are labeled in red.	$\frac{15-20}{7-2} = \frac{-5}{5}$ $m = -1$ They eat 1 ounce per serving.	(0,22) There are 22 ounces to begin.
10. The graph represents the relationship between the total cost of an electrician visit to your home, T, and number of hours he worked, h.  The graph shows a line on a coordinate plane where the horizontal axis is labeled h (hours worked) and the vertical axis is labeled T (total cost). The line starts at the y-intercept (0, 50) and passes through points (1, 150) and (2, 250), which are labeled in red.	$\frac{250-150}{2-1} = \frac{100}{1}$ $m = 100$ The electrician charges \$100 per hour	(0,50) The initial cost of the visit is \$50